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High pressure turbine dirt blocker

Abstract

A dirt blocker including a support structure disposed within a gas turbine engine, the support structure defining an upstream control volume proximate a forward portion of the gas turbine engine and a downstream control volume proximate an aft portion of the gas turbine engine, the downstream control volume being opposite the upstream control volume relative to the support structure, a flow passage formed through the support structure, the flow passage configured to fluidly couple the upstream control volume with the downstream control volume; a radial contact wall formed from the support structure in fluid communication with the upstream control volume, the radial contact wall configured to intercept debris entrained within cooling air within the gas turbine engine; and a stagnation zone fluidly coupled with the flow passage, the stagnation zone configured to reduce momentum of the debris.

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Background/Summary

BACKGROUND

(1) The present disclosure is directed to a system for entrapping entrained debris in the cooling air of a gas turbine engine.

(2) Gas turbine engines are provided with a number of functional sections, including a fan section, a compressor section, a combustion section, and a turbine section. Air and fuel are combusted in the combustion section. The products of the combustion move downstream, and pass over a series of turbine rotors, driving the rotors to provide power.

(3) Numerous components within the gas turbine engine are subject to high levels of heat during operation. As an example, a turbine rotor will have a plurality of turbine blades that are driven by

high temperature products of combustion to rotate and create the power. Cooling fluid, and typically air, is passed within a body of the turbine blades, seals, and vanes to cool the components. (4) The air passing through a gas turbine engine is often subjected to dirt and other impurities. It is desirable that the air utilized for cooling various components be relatively clean. The cooling of the components is through relatively small passages, and the dirt and impurities can clog those small passages.

(5) What is needed is a system that blocks and entrains the dirt to prevent clogging.

SUMMARY

(6) In accordance with the present disclosure, there is provided a dirt blocker comprising a support structure disposed within a gas turbine engine, the support structure defining an upstream control volume proximate a forward portion of the gas turbine engine and a downstream control volume proximate an aft portion of the gas turbine engine, the downstream control volume being opposite the upstream control volume relative to the support structure, a flow passage formed through the support structure, the flow passage configured to fluidly couple the upstream control volume with the downstream control volume; a radial contact wall extending from the support structure in fluid communication with the upstream control volume, the radial contact wall configured to intercept debris entrained within cooling air within the gas turbine engine; and a stagnation zone fluidly coupled with the flow passage, the stagnation zone configured to reduce momentum of the debris.

(7) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the dirt blocker further comprising a debris wall coupled to the support structure proximate an upstream side, the debris wall being fluidly coupled to a bore formed between the radial contact wall and the debris wall, the bore in fluid communication with the flow passage.

(8) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the dirt blocker further comprising an aft facing slot in fluid communication with the bore.

(9) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the aft facing slot is fluidly coupled with the stagnation zone.

(10) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the dirt blocker further comprising an impingement wall coupled to the support structure at a downstream side, the impingement wall configured to intercept debris entrained in the cooling air.

(11) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the impingement wall intersects a centerline of the flow passage, where cooling air flow discharging the flow passage impinges the impingement wall.

(12) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the support structure is disposed within a portion of a high pressure turbine section.

(13) In accordance with the present disclosure, there is provided a dirt blocker for a high pressure turbine comprising a support structure disposed within the high pressure turbine, the support structure defining an upstream control volume proximate a forward portion of the high pressure turbine and a downstream control volume proximate an aft portion of the high pressure turbine, the downstream control volume being opposite the upstream control volume relative to the support structure; a flow passage formed through the support structure, the flow passage configured to fluidly couple the upstream control volume with the downstream control volume; a radial contact wall unitary with the support structure in fluid communication with the upstream control volume, the radial contact wall configured to intercept debris entrained within cooling air within the gas turbine engine; and a stagnation zone fluidly coupled with the flow passage, the stagnation zone configured to reduce momentum of the debris.

(14) A further embodiment of any of the foregoing embodiments may additionally and/or

alternatively include the dirt blocker for a high pressure turbine further comprising a debris wall formed from the support structure proximate an upstream side, the debris wall being fluidly coupled to a bore formed between the radial contact wall and the debris wall, the bore in fluid communication with the flow passage.

(15) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the dirt blocker for a high pressure turbine further comprising an impingement wall coupled to the support structure at a downstream side, the impingement wall configured to intercept debris entrained in the cooling air.

(16) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the impingement wall intersects a centerline of the flow passage, wherein cooling air flow discharging the flow passage impinges the impingement wall.

(17) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the impingement wall comprises multiple, discrete arc segments, individually aligned with the centerline.

(18) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the stagnation zone is located in either the upstream control volume or the downstream control volume.

(19) In accordance with the present disclosure, there is provided a process for removing debris entrained in a gas turbine engine cooling air flow comprising flowing cooling air through the gas turbine engine; bisecting a portion of the gas turbine engine with a support structure disposed within the gas turbine engine, the support structure defining an upstream control volume proximate a forward portion of the gas turbine engine and a downstream control volume proximate an aft portion of the gas turbine engine, the downstream control volume being opposite the upstream control volume relative to the support structure; forming a flow passage through the support structure, the flow passage configured to fluidly couple the upstream control volume with the downstream control volume; forming a radial contact wall unitary with the support structure in fluid communication with the upstream control volume, the radial contact wall configured to intercept debris entrained within cooling air within the high pressure turbine; and forming a stagnation zone fluidly coupled with the flow passage, the stagnation zone configured to reduce momentum of the debris.

(20) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising forming a debris wall unitary with the support structure proximate an upstream side; and fluidly coupling the debris wall to a bore formed between the radial contact wall and the debris wall, the bore in fluid communication with the flow passage.

(21) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising an impingement wall coupled to the support structure at a downstream side; and intercepting debris entrained in the cooling air with the impingement wall.

(22) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process wherein the impingement wall intersects a centerline of the flow passage, wherein cooling air flow discharging the flow passage impinges the impingement wall.

(23) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising locating the stagnation zone in the upstream control volume.

(24) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising locating the stagnation zone in the downstream control volume.

(25) A further embodiment of any of the foregoing embodiments may additionally and/or alternatively include the process further comprising forming raised features as extended material

raised from the upstream side of the support structure adjacent to the flow passage.
(26) Other details of the dirt blocker are set forth in the following detailed description and the accompanying drawings wherein like reference numerals depict like elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic longitudinal sectional view of a gas turbine engine.
- (2) FIG. 2 is a schematic longitudinal sectional view of a portion of the gas turbine engine.
- (3) FIG. 3 is a schematic longitudinal sectional view of a portion of the gas turbine engine with dirt blocker feature.
- (4) FIG. 4 is a schematic longitudinal sectional view of a portion of the gas turbine engine with cooling air control volumes.
- (5) FIG. 5 is a schematic longitudinal sectional view of a portion of the gas turbine engine with dirt blocker feature.
- (6) FIG. 6 is a schematic longitudinal sectional view of a portion of the gas turbine engine with dirt blocker feature.
- (7) FIG. 7 illustrates an enlarged view of an exemplary flow passage.
- (8) FIG. 8 illustrates a process map of an exemplary process.

DETAILED DESCRIPTION

(9) FIG. 1 schematically illustrates a gas turbine engine **20**. The gas turbine engine **20** is disclosed herein as a two-spool turbofan that generally incorporates a fan section **22**, a compressor section **24**, a combustor section **26** and a turbine section **28**. The fan section **22** may include a single-stage fan **42** having a plurality of fan blades **43**. The fan blades **43** may have a fixed stagger angle or may have a variable pitch to direct incoming airflow from an engine inlet. The fan **42** drives air along a bypass flow path B in a bypass duct **13** defined within a housing **15** such as a fan case or nacelle, and also drives air along a core flow path C for compression and communication into the combustor section **26** then expansion through the turbine section **28**. A splitter **29** aft of the fan **42** divides the air between the bypass flow path B and the core flow path C. The housing **15** may surround the fan **42** to establish an outer diameter of the bypass duct **13**. The splitter **29** may establish an inner diameter of the bypass duct **13**. Although depicted as a two-spool turbofan gas turbine engine in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with two-spool turbofans as the teachings may be applied to other types of turbine engines including three-spool architectures.

(10) The exemplary engine **20** generally includes a low speed spool **30** and a high speed spool **32** mounted for rotation about an engine central longitudinal axis A relative to an engine static structure **36** via several bearing systems **38**. It should be understood that various bearing systems **38** at various locations may alternatively or additionally be provided, and the location of bearing systems **38** may be varied as appropriate to the application.

(11) The low speed spool **30** generally includes an inner shaft **40** that interconnects, a first (or low) pressure compressor **44** and a first (or low) pressure turbine **46**. The inner shaft **40** is connected to the fan **42** through a speed change mechanism, which in the exemplary gas turbine engine **20** is illustrated as a geared architecture **48** to drive the fan **42** at a lower speed than the low speed spool **30**. The inner shaft **40** may interconnect the low pressure compressor **44** and low pressure turbine **46** such that the low pressure compressor **44** and low pressure turbine **46** are rotatable at a common speed and in a common direction. In other embodiments, the low pressure turbine **46** drives both the fan **42** and low pressure compressor **44** through the geared architecture **48** such that the fan **42** and low pressure compressor **44** are rotatable at a common speed. Although this application discloses geared architecture **48**, its teaching may benefit direct drive engines having no geared

architecture. The high speed spool **32** includes an outer shaft **50** that interconnects a second (or high) pressure compressor **52** and a second (or high) pressure turbine **54**. A combustor **56** is arranged in the exemplary gas turbine **20** between the high pressure compressor **52** and the high pressure turbine **54**. A mid-turbine frame **57** of the engine static structure **36** may be arranged generally between the high pressure turbine **54** and the low pressure turbine **46**. The mid-turbine frame **57** further supports bearing systems **38** in the turbine section **28**. The inner shaft **40** and the outer shaft **50** are concentric and rotate via bearing systems **38** about the engine central longitudinal axis A which is collinear with their longitudinal axes.

(12) Airflow in the core flow path C is compressed by the low pressure compressor **44** then the high pressure compressor **52**, mixed and burned with fuel in the combustor **56**, then expanded through the high pressure turbine **54** and low pressure turbine **46**. The mid-turbine frame **57** includes airfoils **59** which are in the core flow path C. The turbines **46**, **54** rotationally drive the respective low speed spool **30** and high speed spool **32** in response to the expansion. It will be appreciated that each of the positions of the fan section **22**, compressor section **24**, combustor section **26**, turbine section **28**, and fan drive gear system **48** may be varied. For example, gear system **48** may be located aft of the low pressure compressor, or aft of the combustor section **26** or even aft of turbine section **28**, and fan **42** may be positioned forward or aft of the location of gear system **48**.

(13) The low pressure compressor **44**, high pressure compressor **52**, high pressure turbine **54** and low pressure turbine **46** each include one or more stages having a row of rotatable airfoils. Each stage may include a row of static vanes adjacent the rotatable airfoils. The rotatable airfoils and vanes are schematically indicated at **47** and **49**.

(14) The engine **20** may be a high-bypass geared aircraft engine. The bypass ratio can be greater than or equal to 10.0 and less than or equal to about 18.0, or more narrowly can be less than or equal to 16.0. The geared architecture **48** may be an epicyclic gear train, such as a planetary gear system or a star gear system. The epicyclic gear train may include a sun gear, a ring gear, a plurality of intermediate gears meshing with the sun gear and ring gear, and a carrier that supports the intermediate gears. The sun gear may provide an input to the gear train. The ring gear (e.g., star gear system) or carrier (e.g., planetary gear system) may provide an output of the gear train to drive the fan **42**. A gear reduction ratio may be greater than or equal to 2.3, or more narrowly greater than or equal to 3.0, and in some embodiments the gear reduction ratio is greater than or equal to 3.4. The gear reduction ratio may be less than or equal to 4.0. The fan diameter is significantly larger than that of the low pressure compressor **44**. The low pressure turbine **46** can have a pressure ratio that is greater than or equal to 8.0 and in some embodiments is greater than or equal to 10.0. The low pressure turbine pressure ratio can be less than or equal to 13.0, or more narrowly less than or equal to 12.0. Low pressure turbine **46** pressure ratio is pressure measured prior to an inlet of low pressure turbine **46** as related to the pressure at the outlet of the low pressure turbine **46** prior to an exhaust nozzle. It should be understood, however, that the above parameters are only exemplary of one embodiment of a geared architecture engine and that the present invention is applicable to other gas turbine engines including direct drive turbofans. All of these parameters are measured at the cruise condition described below.

(15) A significant amount of thrust is provided by the bypass flow B due to the high bypass ratio. The fan section **22** of the engine **20** is designed for a particular flight condition—typically cruise at about 0.8 Mach and about 35,000 feet (10,668 meters). The flight condition of 0.8 Mach and 35,000 ft. (10,668 meters), with the engine at its best fuel consumption—also known as “bucket cruise Thrust Specific Fuel Consumption (‘TSFC’)”—is the industry standard parameter of lbf of fuel being burned divided by lbf of thrust the engine produces at that minimum point. The engine parameters described above, and those in the next paragraph are measured at this condition unless otherwise specified.

(16) “Low fan pressure ratio” is the pressure ratio across the fan blade **43** alone, without a Fan Exit

Guide Vane (“FEGV”) system. A distance is established in a radial direction between the inner and outer diameters of the bypass duct **13** at an axial position corresponding to a leading edge of the splitter **29** relative to the engine central longitudinal axis A. The low fan pressure ratio is a span-wise average of the pressure ratios measured across the fan blade **43** alone over radial positions corresponding to the distance. The low fan pressure ratio can be less than or equal to 1.45, or more narrowly greater than or equal to 1.25, such as between 1.30 and 1.40. “Low corrected fan tip speed” is the actual fan tip speed in feet/second divided by an industry standard temperature correction of $[(T_{\text{am}} \text{ } ^\circ \text{R}) / (518.7 \text{ } ^\circ \text{R})]^{.5}$. The “low corrected fan tip speed” can be less than or equal to 1150.0 feet/second (350.5 meters/second), and greater than or equal to 1000.0 feet/second (304.8 meters/second).

(17) Referring also to FIG. 2, details of the cooling air flow through portions of the high pressure turbine structure are shown. Cooling air **60** is shown as line with arrows and dashed lines following a flow path through the various internal components of the gas turbine engine **20**. As the cooling air **60** passes through the vane **62**, the cooling air **60** follows several sharp turns around corners. A dirt purge hole **64** can be located in the vane **62** such that the purged air **66** exits at a location between corners and turns. Debris particles **68** are transported within the cooling air **60**. The large particles **70** and small particles **72** can pass through the relatively large openings in the components. However the large particles **70** can clog certain cooling holes in the components, such as vane cooling holes. Small particles can fill the vane **62** as they become entrapped. The heavier large particles **70** require higher flow rates and velocities to remain being transported. The location of the purge hole **64** in the vane **62** leads to inefficient debris **68** removal. The multiple turns, low Mach number flow, and purge hole **64** location at an angle normal to the cooling air **60** flow direction contributes to the poor removal capability.

(18) Referring also to FIG. 3, an enlarged view of a portion of the high pressure turbine section **54** is shown. The cooling air **60** is shown flowing proximate the case **74** and the air seal **76**. A portion of the support structure **78** is shown with a dirt blocker **80** feature. The support structure **78** can be a part of the supports for the air seals, an extension of the case **74** or other component projection that bisects the cooling air **60** flow as it passes through gas turbine engine **20**. In an exemplary embodiment, the support structure is a wall that creates a pressure difference from one side to another. It is required for the flow metering holes **82** to function. Essentially, support structure **78** is the distinguishing physical boundary that creates the two control volumes, **84** and **86**. It is a pressure vessel.

(19) Referring also to FIG. 4, the cooling air **60** is shown schematically, as a bulk air flow from the front F of the flowpath **60** to the aft rear cavities to cool the high pressure turbine **54** components. The dirt blocker **80** feature in the support structure **78** is in fluid communication with and can bisect the cooling air **60** into an upstream control volume region **84** and a downstream control volume region **86**. An axial station AS can be started from a reference location RL. The upstream control volume **84** can have a high static pressure with very low velocity and Mach number of from about 0.01 to about 0.1. The downstream control volume **86** can have a lower static pressure, very low velocity and Mach number of from about 0.01 to about 0.1. A large percentage of the upstream control volume region **84** can pass to cool the vane **62**. The relative pressures between the two regions can be understood as a static pressure at the downstream control volume to be about 50%-90% of the upstream control volume static pressure.

(20) Referring to FIG. 3, the dirt blocker feature **80** can be configured with a flow passage **82** formed in the support structure **78**. The flow passage **82** can be a radially (relative to the axis A) drilled hole in the support structure **78**. The support structure **78** divides the upstream control volume **84** side **88** of the structure **78** from the downstream control volume **86** side **90** of the structure **78**, such that the cooling air **60** can pass through the flow passage **82** and be in fluid communication. The flow passage **82** can be sized to accelerate the cooling air **60** and entrained debris **68**. The debris **68** entrained in the cooling air **60** can be given momentum sufficient to be

propelled and ricochet.

(21) A first wall or simply a radial contact wall **92** extends radially inward from the support structure **78** on the upstream side **88** and proximate the front F adjacent to the flow passage **82**. The radial contact wall **92** can be unitary (integral) with the support structure **78**. The radial contact wall **92** creates a physical barrier configured to intercept debris **68** and cause impact and reflection (by ricochet) of the first wall **92**. Since the radial contact wall **92** is forward of the flow passage **82**, the radial contact wall **92** receives the initial impact of the debris **68** in the cooling air **60** flow proximate the flow passage **82**. The cooling air **60** flow is forced to flow over the radial contact wall **92**, as depicted by the dashed arrow line. The large particles **70** entrained in the cooling air **60** are also forced to impact the radial contact wall **92**. Smaller particles **72** can follow the cooling air **60** flow over the radial contact wall **92**.

(22) A second wall or simply debris wall **94** extends radially inward from the support structure **78** on the upstream side **88** and aft and adjacent to the flow passage **82** opposite from the radial contact wall **92**. In an alternative embodiment, the first wall **92** and second wall **94** can be a unitary wall with a bore **96** formed to fluidly communicate with the flow passage **82**.

(23) An aft facing slot **98** is formed between the first wall **92** and the second wall **94**. The aft facing slot **98** can be elevated a predetermined distance above the support structure **78** to allow for the accumulation of debris **68** and maintain the slot **98** open to receive cooling air **60** flow. The aft facing slot **98** is situated such that the cooling air **60** flow has to change direction and decelerate. The aft facing slot **98** can be located proximate to a stagnation zone **100** formed by the dirt blocker **80** and the support structure **78**. The stagnation zone **100** can be a location within the upstream control volume **84** (or downstream control volume **86**) proximate the dirt blocker **80** that creates flow deceleration and swirling such that the debris **68** loses momentum and falls out of the cooling air **60** flow and deposits and/or accumulates. The stagnation zone **100** allows for the removal of debris **68** from the cooling air **60** flow.

(24) As seen in FIG. 3 the cooling air **60** can pass through the slot **98** and bore **96** past the flow passage **82** in fluid communication from the upstream control volume **84** into the downstream control volume **86**. The cooling air **60** can flow to the aft sections with less debris **68** entrained within the cooling air **60**.

(25) Referring also to FIG. 5, an exemplary embodiment of the dirt blocker **80** is shown. The dirt blocker **80** can be formed unitary with the support structure **78**. As shown in the other figures, the support structure **78** separates the upstream control volume **84** and downstream control volume **86**. The cooling air **60** is in fluid communication with and passes from the upstream control volume **84** through the flow passage **82** into the downstream control volume **86**. The dirt blocker **80** can be configured from a portion of the support structure **78** being formed as a radial contact wall **92**. The radial contact wall **92** can include a radial orientation relative to the axis A, such that debris **68**, particularly, large particles **70** impinge upon the radial contact wall **92**, lose momentum and drop out of the cooling air **60** flow within the upstream control volume **84**. The flow passage **82** is formed in the support structure **78**, and can be a radially drilled hole (relative to the axis A). The flow passage **82** can be sized to meter the cooling air **60** flow through the support structure **78**. The flow passage can include a predetermined size, for example a size that accelerates the cooling air **60** through the flow passage **82** at a value of from 0.5 Mach number at the upstream control volume **84** to about 1.0 Mach number at the downstream control volume **86**.

(26) The stagnation zone **100** can be located in the downstream control volume **86** side of the support structure **78** proximate the discharge of the flow passage **82**. The stagnation zone **100** can be in fluid communication with the flow passage **82**. The debris **68** that managed to pass through the flow passage **82** and enters the stagnation zone **100** can lose momentum and fall out of the cooling air **60** flow to settle on the case wall **74** as shown. The cooling air **60** can flow to the aft sections without the debris **68**.

(27) Referring also to FIG. 6, an exemplary embodiment of the dirt blocker **80** is shown. The dirt

blocker **80** can include an impingement wall **102**. The impingement wall **102** can extend from or be attached to the support structure **78** on the downstream side of the support structure **78**. The impingement wall **102** can extend proximate the flow passage **82**. The impingement wall **102** can extend across a centerline **104** of the flow passage **82** such that cooling flow **60** exiting the flow passage on the downstream side **90** of the support structure **78**, is forced to impinge the impingement wall **102** to transfer momentum, and change direction from along the centerline **104** to another direction. In an exemplary embodiment, the impingement wall **102** can be multiple, discrete arc segments, individually aligned with the centerline **104**, creating multiple secondary stagnation zones **100**, within the control volume **86**.

(28) The impingement wall **102** is configured to receive debris **68** that is entrained in the cooling air **60** flow exiting the flow passage **82**. The debris **68** that impinges on the impingement wall **102** can be redirected, (ricochet) by the impingement wall **102** and lose momentum. As seen at FIG. **6** the cooling air **60** and entrained debris **68** can be redirected to contact the support structure **78** and lose momentum. The cooling air **60** and entrained debris **68** can be redirected by the impingement wall **102** to contact the downstream side **90** of the support structure **78**. The cooling air **60** and entrained debris **68** can be redirected by the impingement wall **102** to contact the first wall **92** on the downstream side **90** of the support structure **78**.

(29) The impingement wall **102** can be configured to redirect the cooling air **60** flow to the stagnation zone **100**. Multiple ricochets of the debris **68** dissipate the transport energy of the debris **68**. With the momentum and velocity of the debris **68** entrained in the cooling air **60** flow being diminished, the transport mechanism of the debris **68** can be diminished such that the debris **68** can no longer be supported by aerodynamic drag and the debris **68** can drop out of the cooling air **60** flow path. As the debris **68** drops out of the cooling air **60** flow path, the debris **68** can settle on the interior of the case **74** in a location that is low risk, as shown.

(30) FIG. **7** illustrates an enlarged view of the flow passage **82** as previously described. The support structure **78** can include raised features **106**. The raised features **106** can include extended material raised from the upstream side **88** of the support structure **78** adjacent to the flow passage **82**. The raised features **106** can be configured to prevent debris **68** from clogging the flow passage **82**. In an exemplary embodiment, the raised features **106** can be a slotted raised edge of the flow passage **82**.

(31) Referring also to FIG. **8** a process flow diagram of the exemplary embodiment is shown. A process **108** for removing debris entrained in a gas turbine engine cooling air flow is described. The process at step **110** includes flowing cooling air through the gas turbine engine. The process at step **112** includes bisecting a portion of the gas turbine engine with a support structure disposed within the gas turbine engine. The support structure can define an upstream control volume proximate a forward portion of the gas turbine engine and a downstream control volume proximate an aft portion of the gas turbine engine. The downstream control volume being opposite the upstream control volume relative to the support structure. The process can include the step **114** of forming a flow passage through the support structure. The flow passage can be configured to fluidly couple the upstream control volume with the downstream control volume. The process can include the step **116** of forming a radial contact wall unitary with the support structure in fluid communication with the upstream control volume. The radial contact wall can be configured to intercept debris entrained within cooling air within the gas turbine engine. The process can include the step **118** of forming a stagnation zone fluidly coupled with the flow passage. The stagnation zone can be configured to reduce momentum of the debris.

(32) A technical advantage of the disclosed dirt blocker includes reducing cooling hole plugging from debris.

(33) Another technical advantage of the disclosed dirt blocker includes forming the dirt blocker by integration of walls and flow passages designed to remove entrained debris from the cooling air.

(34) Another technical advantage of the disclosed dirt blocker includes a solution based upon physical behaviors of debris, dirt and dust.

(35) Another technical advantage of the disclosed dirt blocker includes the use of walls, turning, impingement barriers and stagnation zones to create multiple places where debris can be blocked.

(36) There has been provided a dirt blocker. While the dirt blocker has been described in the context of specific embodiments thereof, other unforeseen alternatives, modifications, and variations may become apparent to those skilled in the art having read the foregoing description. Accordingly, it is intended to embrace those alternatives, modifications, and variations which fall within the broad scope of the appended claims.

Claims

1. A dirt blocker comprising: a support structure disposed within a gas turbine engine, the gas turbine engine having a central longitudinal axis and comprising a high pressure turbine stage including a high pressure turbine stage vane axially forward of a high pressure turbine stage blade, the support structure disposed radially outward with respect to the central longitudinal axis from the high pressure turbine stage vane, axially aligned with the high pressure turbine stage vane and axially forward of the high pressure turbine stage blade, said support structure defining and dividing an upstream control volume and a downstream control volume, said downstream control volume being opposite said upstream control volume relative to said support structure, wherein upstream and downstream are with respect to cooling air flow direction, a flow passage formed radially through an axially extending portion of said support structure, said flow passage formed between and configured to fluidly couple said upstream control volume with said downstream control volume, all of said upstream control volume being upstream of the flow passage, all of the downstream control volume being downstream of the flow passage; a radial contact wall extending from said support structure within said upstream control volume upstream of the flow passage, said radial contact wall configured to intercept debris entrained within cooling air within said upstream control volume upstream of the flow passage; and a stagnation zone within said upstream control volume fluidly coupled with said flow passage, the stagnation zone being upstream of the flow passage and axially aft of the flow passage, said stagnation zone configured to reduce momentum of said debris upstream of the flow passage.
2. The dirt blocker according to claim 1, further comprising: a debris wall coupled to said support structure proximate an upstream side, the debris wall being upstream of the flow passage, said debris wall being fluidly coupled to a bore formed between said radial contact wall and said debris wall, said bore in fluid communication with said flow passage.
3. The dirt blocker according to claim 2, further comprising an aft facing slot in fluid communication with said bore, the aft facing slot located upstream of the flow passage.
4. The dirt blocker according to claim 3, wherein said aft facing slot is fluidly coupled with said stagnation zone.
5. A dirt blocker for a high pressure turbine comprising: a support structure disposed within said high pressure turbine, said high pressure turbine having a central longitudinal axis and comprising a high pressure turbine stage including a high pressure turbine stage vane axially forward of a high pressure turbine stage blade, the support structure disposed radially outward with respect to the central longitudinal axis from the high pressure turbine stage vane, axially aligned with the high pressure turbine stage vane and axially forward of the high pressure turbine stage blade, said support structure defining an upstream control volume and a downstream control volume, said downstream control volume being opposite said upstream control volume relative to said support structure, said support structure dividing the upstream control volume from the downstream control volume, wherein upstream and downstream are with respect to cooling air flow direction; a flow passage formed radially through an axially extending portion of said support structure, said flow passage configured to fluidly couple said upstream control volume with said downstream control volume, the flow passage formed between the upstream control volume and the downstream

control volume, all of the upstream control volume being upstream of the flow passage, all of the downstream control volume being downstream of the flow passage; a radial contact wall unitary with said support structure within said upstream control volume upstream of the flow passage, said radial contact wall configured to intercept debris entrained within cooling air upstream of the flow passage within the upstream control volume of said high pressure turbine before passing through the flow passage; and a stagnation zone fluidly coupled with said flow passage, the stagnation zone being upstream of the flow passage and axially aft of the flow passage, said stagnation zone configured to reduce momentum of said debris.

6. The dirt blocker for a high pressure turbine according to claim 5, further comprising: a debris wall formed from said support structure proximate an upstream side, said debris wall being fluidly coupled to a bore formed between said radial contact wall and said debris wall, the debris wall being upstream of the flow passage, said bore in fluid communication with said flow passage.

7. A process for removing debris entrained in a gas turbine engine cooling air flow, the gas turbine engine comprising: a central longitudinal axis; a high pressure turbine stage including a high pressure turbine stage vane axially forward of a high pressure turbine stage blade; a support structure disposed within the gas turbine engine radially outward with respect to the central longitudinal axis from the high pressure turbine stage vane, axially aligned with the high pressure turbine stage vane and axially forward of the high pressure turbine stage blade, said support structure defining and dividing an upstream control volume and a downstream control volume, said downstream control volume being opposite said upstream control volume relative to said support structure, wherein upstream and downstream are with respect to cooling air flow direction, a flow passage formed radially through an axially extending portion of said support structure, said flow passage formed between and configured to fluidly couple said upstream control volume with said downstream control volume, all of said upstream control volume being upstream of the flow passage, all of the downstream control volume being downstream of the flow passage; a radial contact wall extending from said support structure within said upstream control volume upstream of the flow passage, said radial contact wall configured to intercept debris entrained within cooling air within said upstream control volume upstream of the flow passage; and a stagnation zone within said upstream control volume fluidly coupled with said flow passage, the stagnation zone being upstream of the flow passage and axially aft of the flow passage, said stagnation zone configured to reduce momentum of said debris upstream of the flow passage; and the process for removing debris entrained in the gas turbine engine cooling air flow comprising: flowing the cooling air through the gas turbine engine; bisecting a portion of the high pressure turbine stage of said gas turbine engine with the support structure disposed radially outward with respect to the central longitudinal axis from the high pressure turbine stage vane, axially aligned with the high pressure turbine stage vane and axially forward of the high pressure turbine stage blade within said gas turbine engine; forming the radial contact wall unitary with said support structure in fluid communication with said upstream control volume; flowing the cooling air into the upstream control volume, said radial contact wall configured to intercept debris entrained within the cooling air; forming the stagnation zone within the upstream control volume fluidly coupled with said flow passage, said stagnation zone configured to reduce momentum of said debris; and flowing the cooling air through the flow passage through said support structure, said flow passage fluidly coupling said upstream control volume with said downstream control volume.

8. The process of claim 7, further comprising: forming a debris wall unitary with said support structure proximate an upstream side; and fluidly coupling said debris wall to a bore formed between said radial contact wall and said debris wall, said bore in fluid communication with said flow passage.
